

Nice Properties of the Local Outlier Factor

- ❑ For an object deep within a consistent cluster, the local outlier factor is close to 1
- ❑ For an object o , the meaning of $LOF(o)$ is easy to understand

$$direct_{min}(o) = \min\{reachdist_k(o' \leftarrow o) | o' \in N_k(o)\}$$

$$direct_{max}(o) = \max\{reachdist_k(o' \leftarrow o) | o' \in N_k(o)\}$$

$$indirect_{min}(o) = \min\{reachdist_k(o'' \leftarrow o') | o' \in N_k(o) \text{ and } o'' \in N_k(o')\}$$

$$indirect_{max}(o) = \max\{reachdist_k(o'' \leftarrow o') | o' \in N_k(o) \text{ and } o'' \in N_k(o')\}$$

LOF(o) is bounded:

$$\frac{direct_{min}(o)}{indirect_{max}(o)} \leq LOF(o) \leq \frac{direct_{max}(o)}{indirect_{min}(o)}$$

